

Reproduction of KellerJordan/cifar10-airbench: airbench94_muon CIFAR-10 speedrun

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Submitted by K-Veritas Team

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Independent reproduction · Artifact evaluation · Machine learning

Abstract

Ran airbench94_muon.py unmodified at the pinned commit: one compile warmup run followed by the script's default 200 independent CIFAR-10 training runs (8 epochs each, Muon optimizer, torch.compile max-autotune), each evaluated with test-time augmentation; the script reports the mean and standard deviation of test accuracy over the 200 runs. CIFAR-10 downloaded by torchvision. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	CIFAR-10 airbench94 (200 ru...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: CIFAR-10 airbench94 (200 runs)

Run	Metric	Final value	Step
1	tta_val_acc	0.9417	199
1	time_seconds	9.4859	199

Run	Inline claim	Value	Phase
1	mean_tta_val_acc	0.94	
1	std_tta_val_acc	0.0013	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	2083.8 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	ab1e6d396ccb7f83f2286108042f0edad15c2a7dfe266efe50fc8e05e1d434f9	7.0 MB
r1.kvbundle.zip	838ea0df39f795e3f42966c506a0e4186094e169a537719c970060d92245fadf	63 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2404.00498>

Reproduction of: <https://github.com/KellerJordan/cifar10-airbench/tree/4c1b6d1e3889b037efadcfd5c0ea65b246592362>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: cifar-10, speedrun, muon, image-classification.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (CIFAR-10 airbench94 (200 runs)), kv:2610.00011v1/r1. Data hash 8ac04041ba4715ba478a7ae129b69d13a010bf7bc1053910fed09081f2454932.
<https://kveritas.org/records/2610.00011v1/r1.pdf>